Q1.

A student investigates what happens when light travels from air to glass.

Figure 2 shows some of the apparatus used in the investigation.

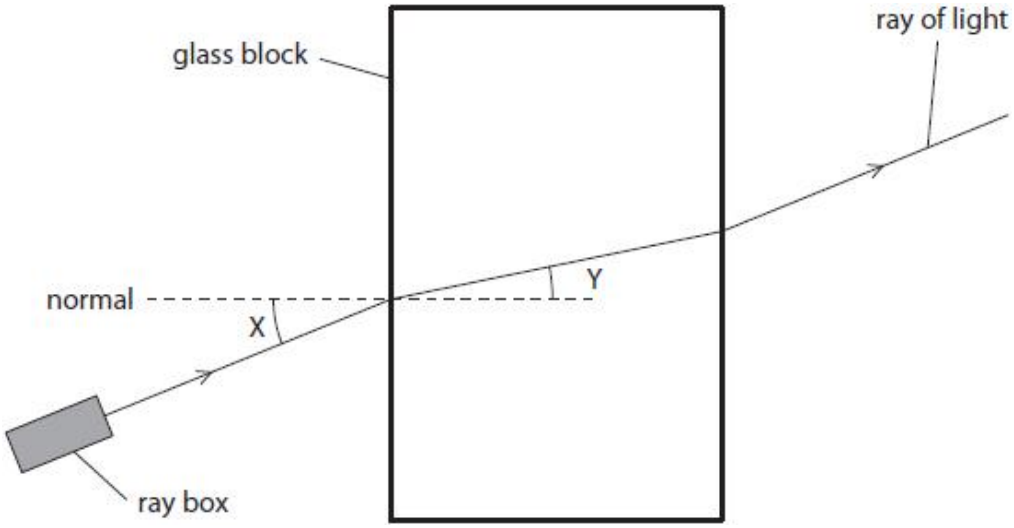

Figure 2

(i) In Figure 2, angle Y is the angle of

(1)

- ☒ **A** deflection
- ☒ **B** incidence
- ☒ **C** reflection
- ☒ **D** refraction

(ii) Figure 3 is a graph of the student's results.

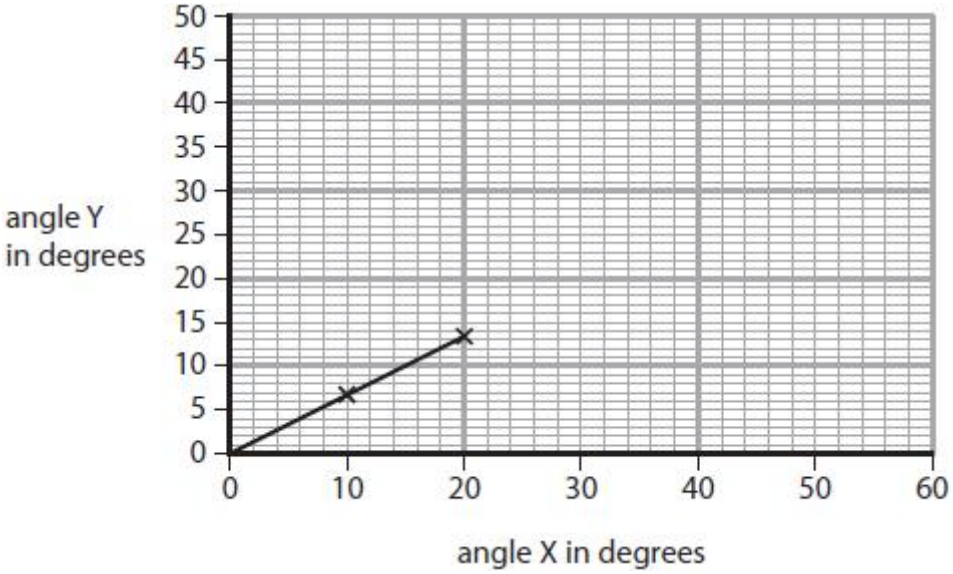

Figure 3

Use the graph to calculate a value for

$$\frac{\text{angle Y}}{\text{angle X}}$$

(2)

$\frac{\text{angle Y}}{\text{angle X}} =$ .....

(iii) The student concludes that angle Y is directly proportional to angle X.

Explain what the student must do to test this conclusion in more detail.

(3)

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(Total for question = 6 marks)

Q2.

Diagram 1 shows a glass prism which can be used to turn an image the right way up.

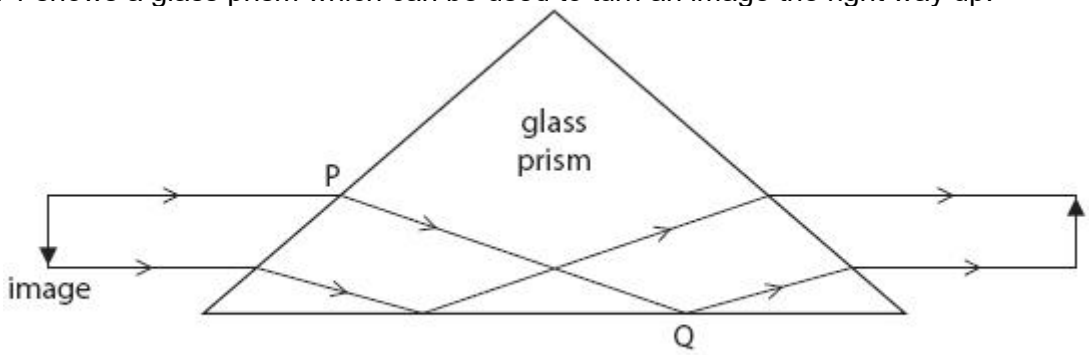

Diagram 1

(a) (i) In diagram 1, total internal reflection occurs at Q.

Explain why total internal reflection occurs at Q.

(2)

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(ii) The way in which the light changes direction at P is shown in diagram 2.

Mark on the diagram (*i*) for the angle of incidence and (*r*) for the angle of refraction for the ray of light shown.

(2)

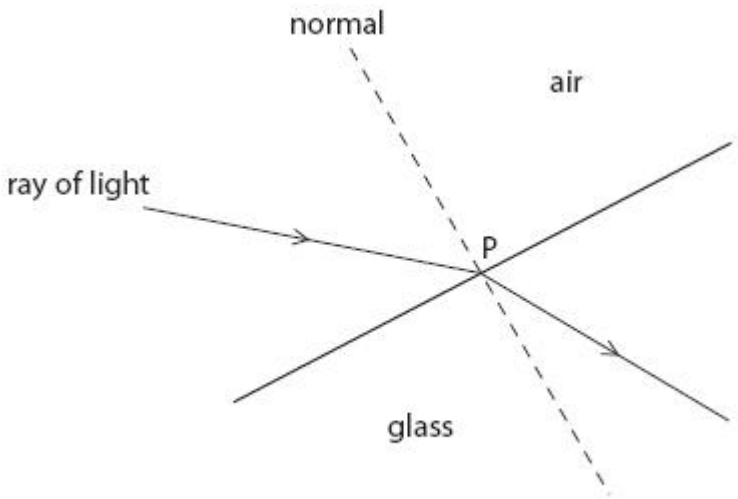

Diagram 2

(iii) Which of these is correct for the light as it enters the prism at P?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** frequency decreases

☐ **B** frequency increases

☐ **C** speed decreases

☐ **D** speed increases

(b) Light waves and sound waves are both used in the diagnosis and treatment of medical conditions

(i) A doctor uses an endoscope to look inside the body of a patient.
Explain how optical fibres are used in endoscopes.
You may draw a labelled diagram to help with your answer.

(3)

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(ii) Describe how ultrasound can be used as a medical treatment for illness or injury.

(2)

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(Total for Question = 10 marks)

Q3.

Radio waves from Jupiter take 40 minutes to reach Earth.

Light waves from the Sun take 8 minutes to reach Earth.

Calculate how many times further it is from Earth to Jupiter than from Earth to the Sun.

State the property of electromagnetic radiation that is used in your answer.

(2)

..... times

property .....

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(Total for question = 2 marks)

Q4.

Figure 2 shows a converging lens used to produce a real image on a screen.

The apparatus is used to find the focal length, *f*, of the lens.

The screen is moved to give a clear image of the object.

The distance from the **object** to the **lens**, *a* = 20 cm.

The distance from the **image** to the **lens**, *b* = 40 cm.

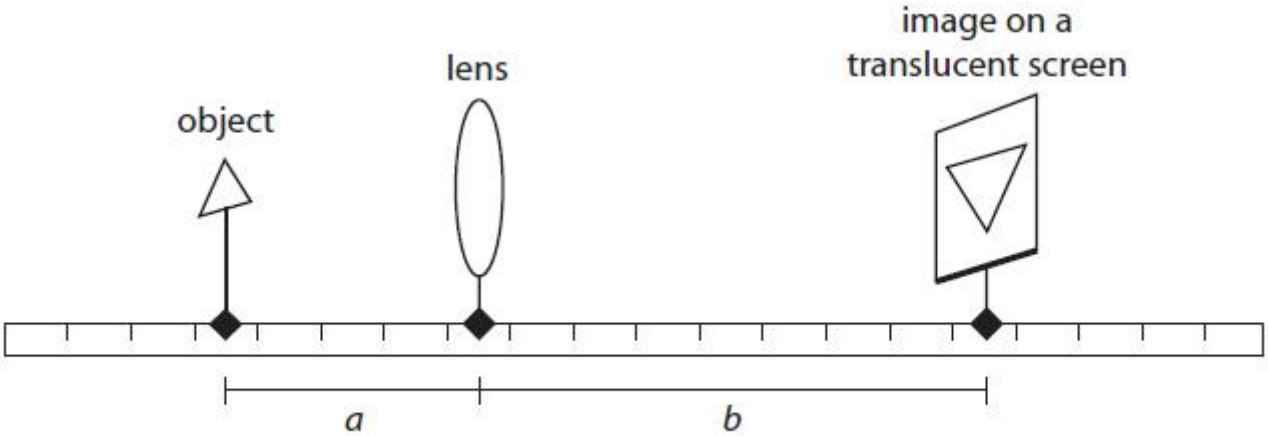

Figure 2

Calculate the focal length, *f*, of the lens.

Use the equation

$$\frac{1}{f} = \frac{a+b}{a \times b}$$

(3)

focal length *f*= ..... cm

(Total for question = 3 marks)

Q5.

The Asteroid Belt is part of our Solar System.

Vesta is an asteroid in the Asteroid Belt.

The distance between Vesta and the Sun is 2.4 AU.

1 AU is the distance between the Earth and the Sun.

The intensity of the Sun's radiation reaching the Earth is 1400 W/m^2.

1 W = 1 J/s

The intensity of the Sun's radiation at a distance, *d*, from the Sun is given by the equation

$$\text{intensity} = \frac{K}{(d)^2}$$

where *K* always has the same value.

(i) State the unit of *K*.

(1)

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(ii) Calculate the intensity of the radiation from the Sun at Vesta.

(3)

intensity = ... W/m^2

(Total for question = 4 marks)

Q6.

This question is about different types of electromagnetic radiation.

The human eye can detect a range of wavelengths, from violet at about 400 nm to red at about 700 nm. Figure 8 shows the relative sensitivity of the human eye to different wavelengths when inside a dark cave.

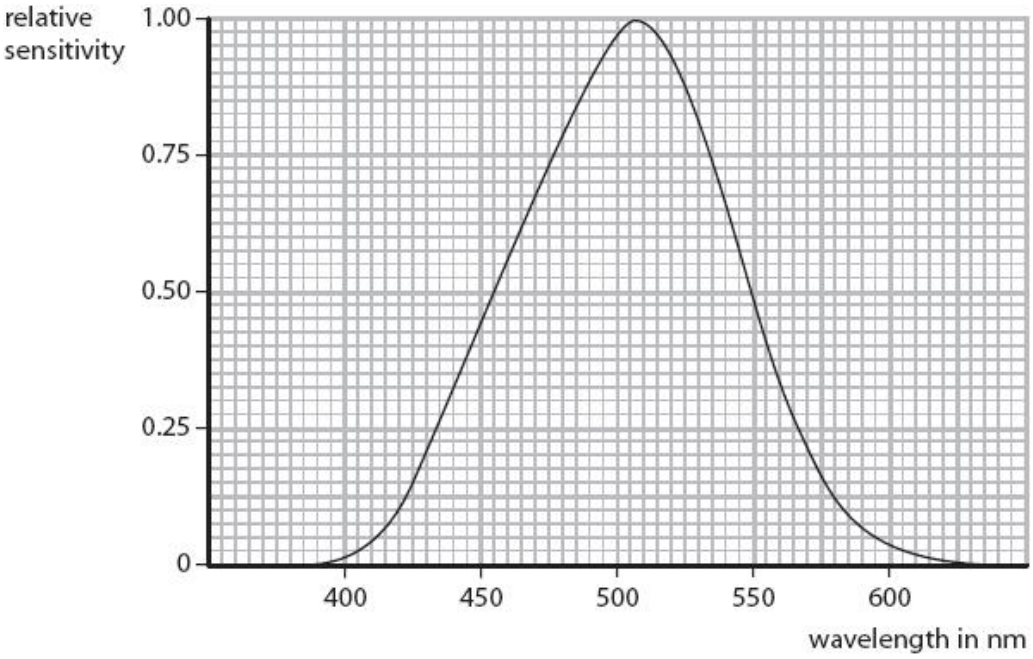

Figure 8

(i) Use the information in Figure 8 to determine the wavelength to which the eye is most sensitive inside the cave.

(1)

wavelength = ... nm

(ii) Calculate the frequency to which this wavelength corresponds. Give the unit.

The speed of light in air = 3.0×10^8 m/s.

(3)

frequency = ... unit .....

(iii) Using information from Figure 8, describe how well the eye detects different colours of the visible spectrum inside the cave.

(3)

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(Total for question = 7 marks)

Q7.

This question is about waves in the electromagnetic (e.m.) spectrum.

The potential danger associated with the waves of the e.m. spectrum increases as

(1)

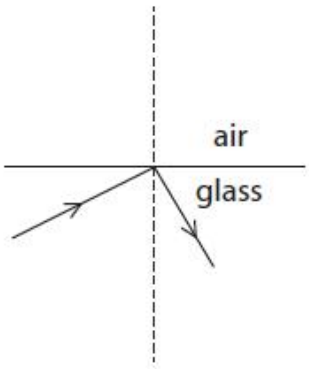
- ☒ **A** frequency decreases
- ☒ **B** frequency increases
- ☒ **C** velocity decreases
- ☒ **D** velocity increases

(Total for question = 1 mark)

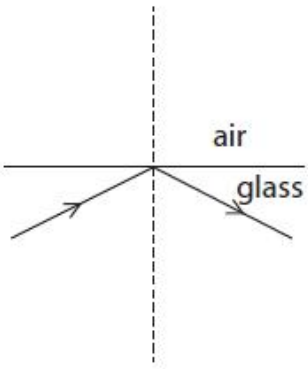
Q8.

Which ray diagram shows total internal reflection at an air and glass boundary?

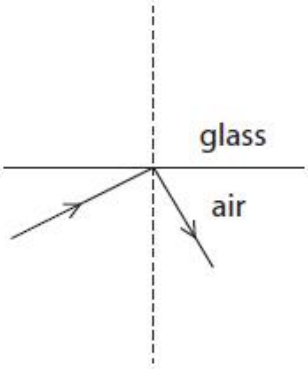
(1)



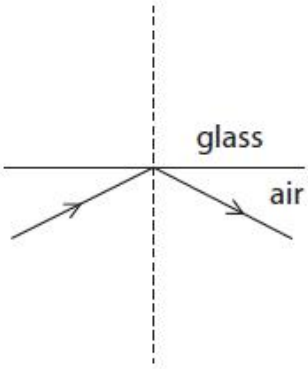
☒ **A**



☒ **B**



☒ **C**



☒ **D**

(Total for question = 1 mark)

Q9.

Radio waves and gamma radiation are at opposite ends of the electromagnetic spectrum.

Compare how these two electromagnetic radiations are produced.

(6)

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(Total for question = 6 marks)

Q10.

* Gamma radiation is produced by radioactive decay.

Alpha radiation and beta radiation are also produced by radioactive decay.

Compare the processes of alpha decay and beta decay.

Your answer should include what each radiation is and what effect each decay has on the original nucleus.

(6)

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(Total for question = 6 marks)

Mark Scheme

Q1.

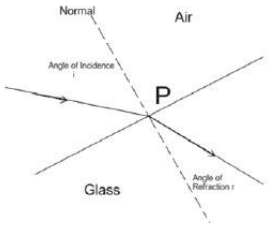
Question Number	Answer	Mark
(i)	**D** refraction **is the only correct answer** *A 'deflection' is an incorrect distracting description* *B 'incidence' is incorrect, that would be angle X* *C 'reflection' is incorrect, no reflection being shown in the diagram*	(1) AO 1 1

Question Number	Answer	Additional guidance	Mark
(ii)	any pair of coordinates selected from the line (1) in range → 0.6(0) to 0.7(0) (1)	e.g. 20 and (13 or 14) or 10 and (6 or 7) ignore any units given award full marks for a correct answer without working	(2) AO 2 1

Question Number	Answer	Additional guidance	Mark
(iii)	an explanation linking: repeat (1) different angles / more values of X (1) for larger angles / values of X (1)	 allow 'more measurements' / 'repeat experiment' / collect more data > 20°	(3) AO 3 3a

Q2.

	Answer	Acceptable answers	Mark
(a)(i)	An explanation linking:	Angle in air cannot be greater	**(2)**

	Angle (of incidence) in glass (1) greater than critical angle / 42° (1)	than 90° for 1 mark Glass has a higher refractive index than air for 1 mark	
(a)(ii)	 angle i from normal in air (1) angle r from normal in glass (1)	accept for 1 mark angle i in air <u>and</u> angle r in glass/ <u>both</u> angles measured from normal	**(2)**
(a)(iii)	☒ **C** speed decreases		**(1)**
(b)(i)	An explanation linking any three of the following: (Optical fibres) bend (1) some fibres carry light to the inside of the patient (1) some fibres transmit the reflected light (1) light passes up/down fibres by TIR (1) light is reflected inside the patient (1) image is analysed by computer (1)	Accept suitable labelling on a Diagram Image projected on a screen	**(3)**
(b)(ii)	Either Breaks/blasts/smashes (1) Kidney stones (1) or Energy absorbed (1) to help repair muscle tissue (1) or Use of gel (1) to prevent loss of intensity (1) or	bruising/clots increases blood flow Allow (1) mark for suitable diagnosis e.g. prenatal scan	**(2)**

Q3.

SSQ NO:	Answer	Additional guidance	Mark
	(Jupiter is) 5 times (further away) (1) radio waves and light waves travel at the same speed (in space) (1)	All electromagnetic (EM) waves travel at the same speed accept attempt to use consistent speed (of light) to calculate two distances	**(2)**

Q4.

Question Number	Answer	Additional guidance	Mark
	substitution (1) $\frac{1}{f} = \frac{20+40}{20\times40}$ evaluation of $\frac{1}{f}$ (1) 0.075 evaluation of f (1) 13 (cm)	accept $\frac{1}{f} = \frac{60}{800}$ OR $\frac{1}{f} = \frac{3}{40}$ accept 13.3 (cm) 13.33 (cm) power of ten error on 13 gains 2 marks 0.075 on answer line gains 2 marks power of ten error on 0.075 gains 1 mark $\frac{40}{3}$ scores 2 marks award full marks for correct answer without working	**(3)** **AO2**

Q5.

	Answer	Additional guidance	Mark
(i)	W (1)	accept watt(s) J/s do **NOT** accept Ws W/s watt(s) per second	**(1)** **AO2**

	Answer	Additional guidance	Mark
(ii)	evaluation of K (1) $K = 1^{(2)} \times 1400$ substitution (1) (intensity =) $\frac{1400}{2.4^2}$ evaluation (1) 240 (W/m^2)	 1400 accept for one mark intensity = $\frac{K}{2.4^2}$ allow numbers that round to 240 e.g. 243.05 accept answers in terms of ratios award full marks for correct answer without working.	(3) AO2

Q6.

Question number	Answer	Additional guidance	Marks
(i)	Accept any answer between 502 and 510 nm inclusive		1

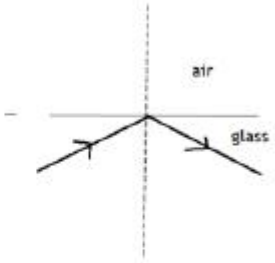
Question number	Answer	Additional guidance	Marks
(ii)	Substitution (1) frequency = 3.0×10^{8} / (505×10^{-9}) Evaluation (1) $=5.9 \times 10^{14}$ Unit mark (1) Hz	 accept answers in range 5.88×10^{14} to 5.98×10^{14} Accept hertz / s^{-1}	(3)

Question number	Answer	Additional guidance	Marks
(iii)	An answer that combines three points of interpretation/evaluation to provide a logical description (3) Yellow or green or blue detected/seen the best/most sensitive to / highest relative sensitivity (1) Red or orange not detected/not seen (so well)/least sensitive to/lowest relative sensitivity (1) violet or indigo not detected/not seen (so well)/less sensitive to (1)	 about 500 nm more than 550 nm less than 450 nm 2 max if only wavelength instead of colours	(3)

Q7.

Question Number	Answer	Mark
	B frequency increases ***A*** *is not correct because the danger does not increase with decreasing frequency* ***C*** *is not correct because all waves in the e-m spectrum have the same velocity* ***D*** *is not correct because all waves in the e-m spectrum have the same velocity*	**(1)** **AO1**

Q8.

Qustion Number	Answer	Mark
	B  **Once DTP redraw image, crop and insert here.** ***A*** *is not correct because the angle of incidence is not equal to the angle of reflection* ***C*** *is not correct because glass is more optically dense than air* ***D*** *is not correct because glass is more optically dense than air*	**(1)** **AO1**

Q9.

Question Number	Answer	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 strand 1 (6 marks)** • radio waves are (often) produced intentionally (by humans) • gamma rays are (often) produced spontaneously / randomly • radio waves are produced by (free) electrons • radio waves are produced by oscillating (free) electrons / alternating current (ac) • radio waves are produced in electrical circuits / aerials • gamma rays may result from radioactive decay • gamma rays produced in the nucleus • gamma rays produced by energy changes / rearrangement in the nucleus • gamma rays produced to stabilise the nucleus • gamma rays produced in annihilations (PET scanning etc) • gamma rays may be produced as a result of (nuclear) fission or fusion	**(6)**

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Summary for guidance			
Level	**Mark**	**Additional Guidance**	**General additional guidance – the decision within levels** e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated fact(s) about one radiation	Possible candidate responses gamma rays are (often) produced spontaneously / randomly
Level 2	3–4	Additional guidance Some understanding shown i.e. a limited comparison made including some facts about the production of each radiation OR more detailed facts given about the production of one of them	Possible candidate responses radio waves produced in wires and gamma produced in nucleus radio waves produced by AC in wires
Level 3	5–6	Additional guidance Understanding is detailed and fully developed. detailed comparison made with linked facts about the production of each (one radiation may have significantly more detail than the other but both should feature for level 3)	Possible candidate responses radio waves produced by electrons oscillating in wires; gamma produced by annihilation of electrons interacting with positrons

Q10.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive, and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. AO1 1 **(6 marks)** **alpha** • a particle (not a wave) • made up of 4 particles • helium nucleus • has a positive charge • when emitted by a nucleus, atomic number goes down by 2 • mass number goes down by 4 **beta** • a particle (not a wave) • made up of 1 particle • electron (or positron) • has a negative charge • when emitted, atomic number goes up by 1 • mass number does not change Ignore references to range, penetration, ionisation.	**(6)** **AO1**

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated facts	Possible candidate responses A beta particle is an electron. An alpha particle is a helium nucleus
Level 2	3–4	Additional guidance effect of alpha and beta decay or nature and effect of alpha or beta	Possible candidate responses A beta particle is an electron. When emitted the mass number doesn't change but atomic number goes up by one
Level 3	5–6	Additional guidance detailed comparison that includes nature of alpha and nature of beta and effect of either alpha or beta OR effect of alpha and beta and nature of either alpha or beta	Possible candidate responses Alpha particle is a helium nucleus AND A beta particle is an electron. When emitted the mass number doesn't change but atomic number goes up by one